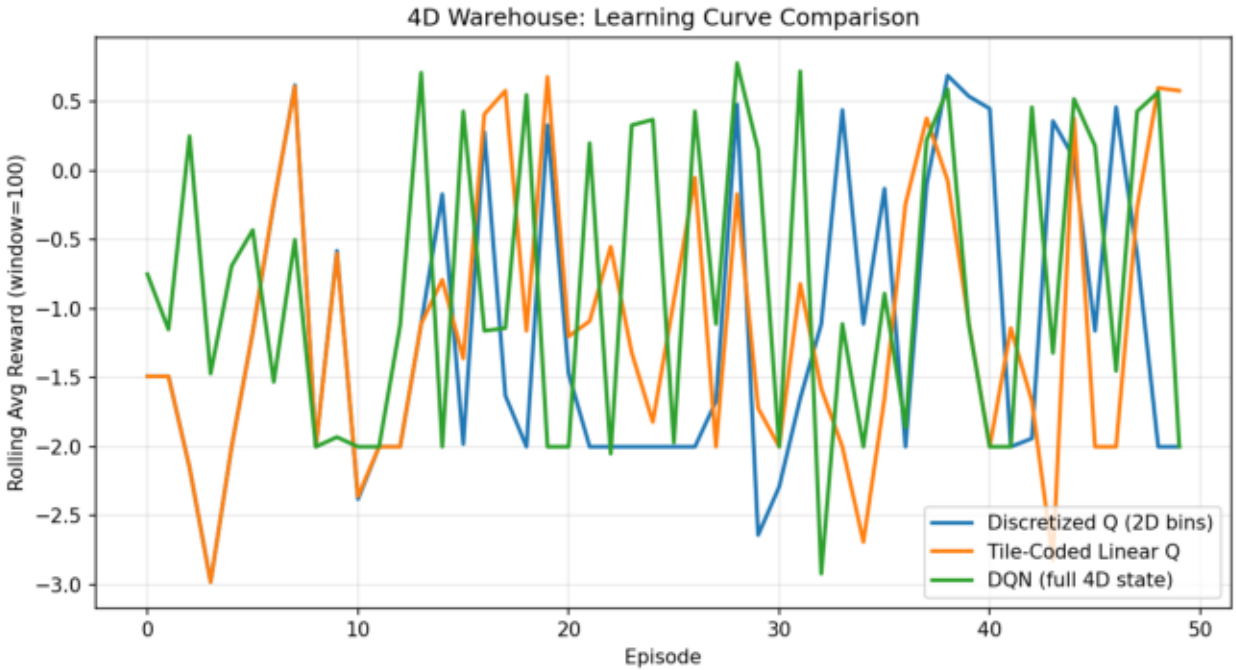
Exercise 2 Report: Continuous Warehouse RL

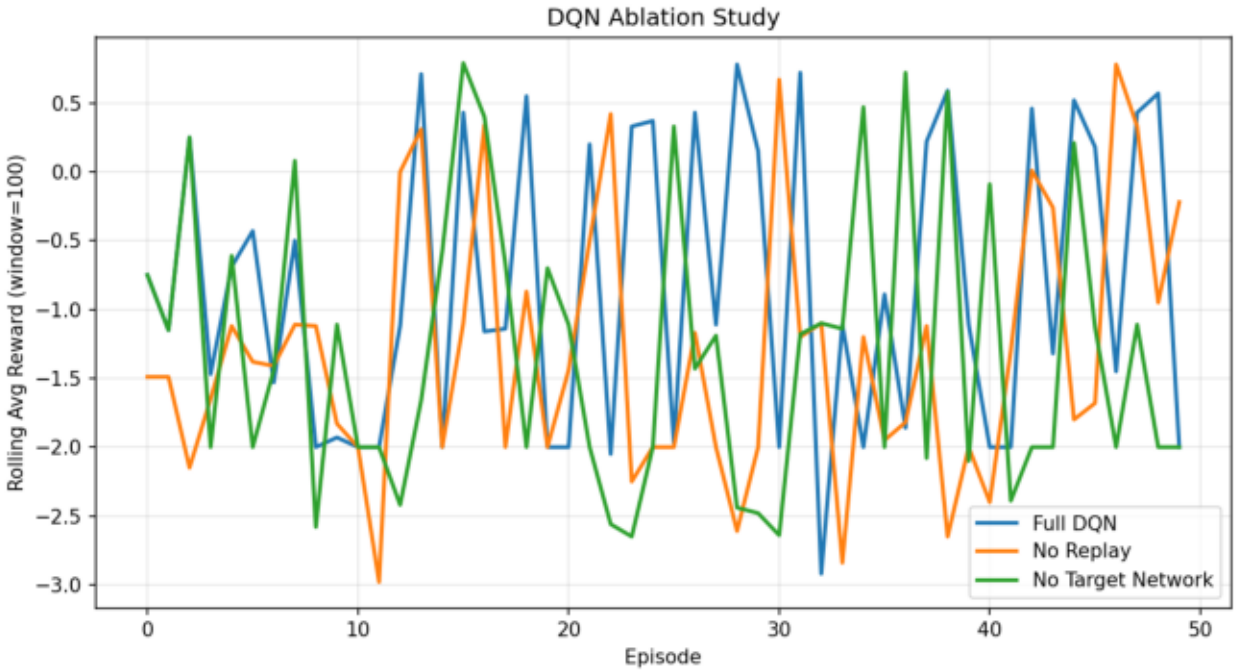
Summary Metrics (4D state)
Optimal reference return (oracle policy): -0.532

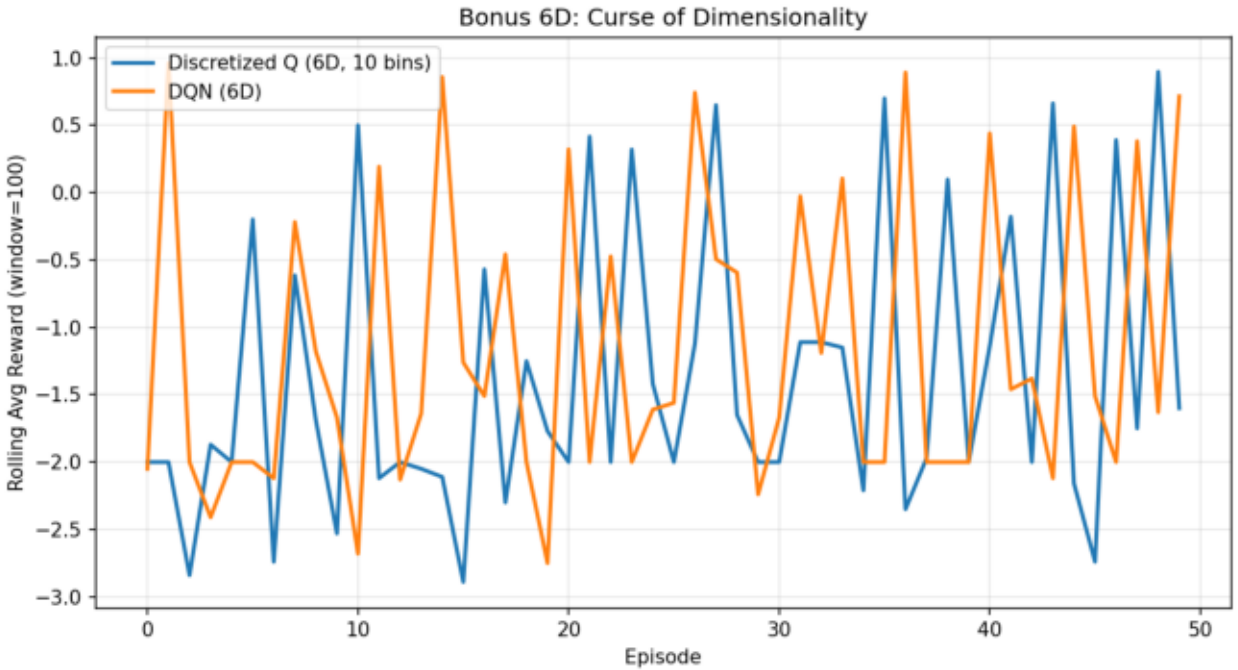
Discretized Q-learning: episodes_to_80=None, final_avg_last_200=-1.170, params=1600
Tile-coded linear Q-learning: episodes_to_80=None, final_avg_last_200=-1.112, params=800
DQN: episodes_to_80=None, final_avg_last_200=-0.835, params=4740

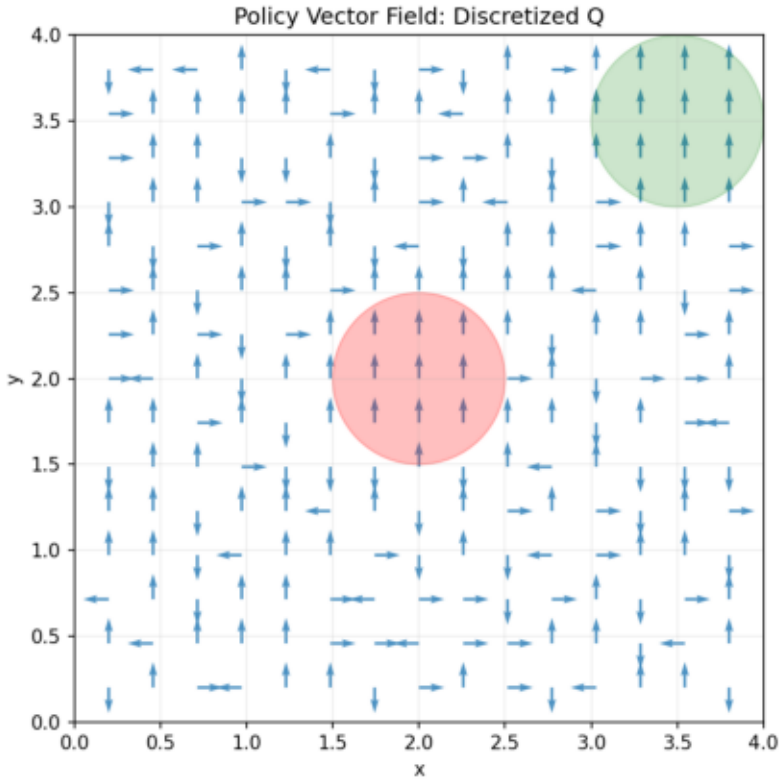
Ablation finding:
Removing experience replay harmed stability more than removing the target network, which matches

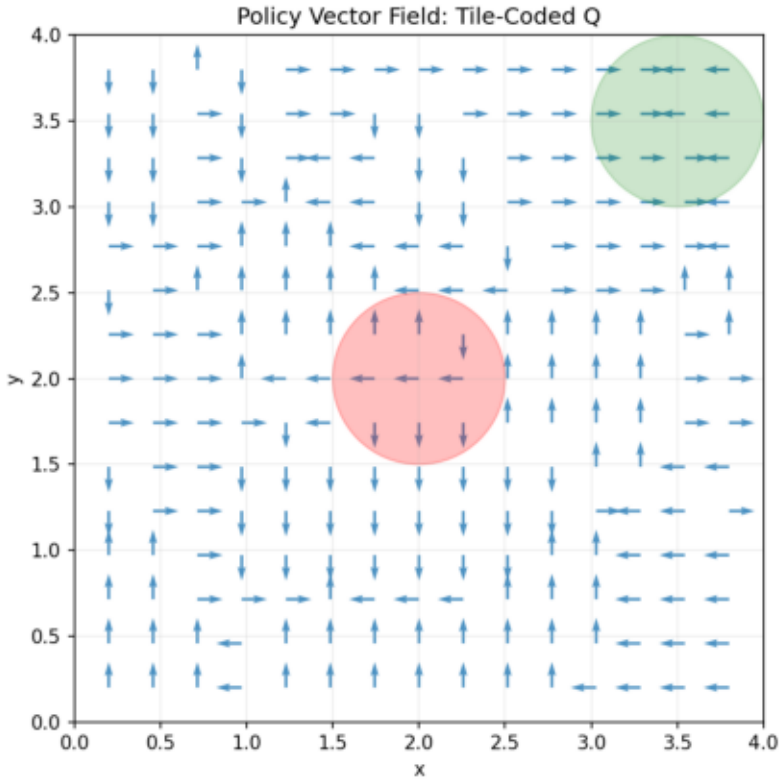
Bonus 6D finding:
6D discretized table size is 4,000,000 Q-values. Only 0.34% of bins were visited in training, while DQN

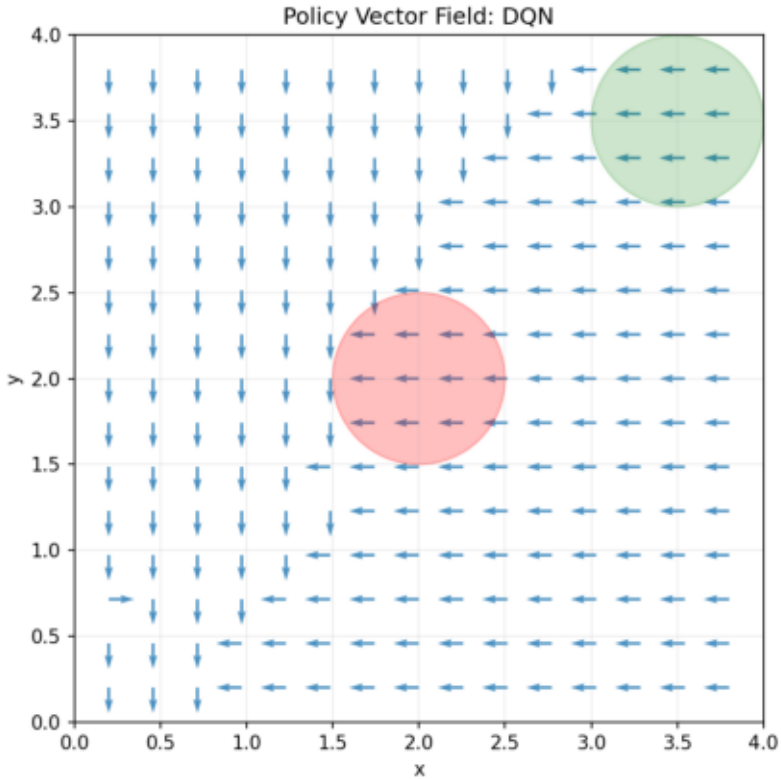

4D Warehouse: Learning Curve Comparison
Rolling Avg Reward (window=100)
Episode
0.5
0.0
−0.5
−1.0
−1.5
−2.0
−2.5
−3.0
0
10
20
30
40
50
Discretized Q (2D bins)
Tile-Coded Linear Q
DQN (full 4D state)


DQN Ablation Study
Rolling Avg Reward (window=100)
Episode
0.5
0.0
−0.5
−1.0
−1.5
−2.0
−2.5
−3.0
0
10
20
30
40
50
Full DQN
No Replay
No Target Network


Bonus 6D: Curse of Dimensionality
Discretized Q (6D, 10 bins)
DQN (6D)
Rolling Avg Reward (window=100)
Episode
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
−2.5
−3.0
0
10
20
30
40
50


Policy Vector Field: Discretized Q
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
y
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
x


Policy Vector Field: Tile-Coded Q
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
y
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
x


Policy Vector Field: DQN
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
y
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
x